

CSCE 753 COMPUTER VISION · MODULE 12 · LECTURE NOTES

Evaluation and Failure

What a vision system can be trusted to do.

MODULE OUTCOMES

- Explain distribution shift and its kinds.
- Design an evaluation that measures robustness.
- Explain shortcut learning and detect it.
- Explain adversarial examples and their practical relevance.
- State a scoped trust claim for a vision system.

1 Distribution shift

Kind	What changes	Example in vision
Covariate shift	The input distribution; the labelling rule is unchanged.	A new camera, new lighting, a new installation site, a different season. Every new deployment is one of these.
Label shift	The class frequencies.	A condition that was rare in the training hospital is common in the deployment one, so the calibrated thresholds are wrong.
Concept shift	The relationship between input and label itself.	What counts as 'damaged' or 'defective' is redefined by the people using the system.
Subpopulation shift	A group that was under-represented in training becomes a larger share of the input.	Skin tones, body types, vehicle types, regional signage. This is the kind that causes harm, and aggregate accuracy conceals it by construction.
Adversarial shift	Inputs deliberately chosen to fail.	A targeted attack — see §3 for the proportionate response.

THE BENCHMARK GAP IS MEASURED, NOT HYPOTHETICAL

When researchers carefully collected a new ImageNet test set following the original protocol as closely as they could, every model dropped by a substantial and consistent margin. Not because the new set was harder in any identifiable way, but because the original had been selected against by a decade of model development.

Under corruptions — blur, Gaussian noise, simulated weather, JPEG compression — accuracy falls far more, and the size of the fall is not predicted by clean accuracy, which means it has to be measured separately.

And rankings change. The best model on the clean benchmark is frequently not the best under shift — so selecting a model on clean accuracy can select the wrong model for deployment, which is a stronger and more actionable claim than 'accuracy drops'.

Which means your test set measures your test set. It is a lower bound on the difficulty of the real problem and not an estimate of deployed performance — and this is a research finding with numbers attached, not a cautious hedge. **It is CSCE 633 Module 12's distribution-shift section, with the vision literature's specific measurements behind it.**

2 Shortcut learning

WHAT HAPPENED, IN REAL DEPLOYED AND PUBLISHED SYSTEMS:

- a pneumonia classifier that read the HOSPITAL's scanner signature from image metadata burned into the pixels, because prevalence differed between hospitals
- a skin-lesion classifier that used the RULER that clinicians place beside lesions they already suspect are malignant
- a cow detector that relied on GRASS, and failed on photographs of cows on a beach
- a "COVID from chest X-ray" model that used patient POSITIONING, which differed between the wards where positive and negative cases were imaged

EVERY ONE OF THESE WORKED ON THE TEST SET. The shortcut was genuinely predictive there, because the test set came from the same collection process as the training set.

HOW TO FIND YOURS

- occlude parts of the input; if accuracy survives with the object removed, the object was not being used
- evaluate PER SOURCE -- per hospital, camera, site, photographer -- and look for a model that is excellent within each source and poor across them
- look at saliency maps, with suspicion: they are unreliable, but a map pointing at a corner watermark is a signal worth following
- test on data YOU collected, differently, yourself
- and show the failure gallery to a domain expert

A shortcut is not a bug in the model. It is a real correlation in your data that you did not intend to teach, and the model found it because it was easier to use than the feature you had in mind — which is exactly what an optimiser should do. **So the fix is in the data and the evaluation, not in the architecture,** and **per-source evaluation is the cheapest reliable detector:** it costs a groupby and it catches most of the documented cases above.

3 Adversarial examples

ADVERSARIAL EXAMPLES, IN PROPORTION

An imperceptible, carefully computed perturbation of an image flips a confident prediction to a different confident prediction. This is a real, reproducible property of essentially every vision model, including large and modern ones, and it has resisted a decade of defences.

What it establishes: the decision boundary lies far closer to typical inputs than the model's confidence suggests. That is a statement about calibration and about the geometry of the learned function, and it is genuinely important — it means confidence cannot be read as distance from the boundary.

What it does not establish: that the model is useless, or that robustness to attack is the main robustness problem. **The overwhelming majority of deployment failures are ordinary distribution shift and shortcut learning, not attacks** — and for most systems that is where the effort belongs.

When it does matter: when an adversary benefits from a misprediction and can influence the input. **Physical attacks exist** — printed patterns that defeat detectors, stickers that alter sign classification — **so the assessment should be by threat model rather than by whether the phenomenon is real. It is real; whether it is your problem is a separate question, and conflating the two leads both to complacency and to misdirected effort.**

4 What a real evaluation contains

- **A test set collected separately from the training data** — a different day, a different place, a different device, a different operator. **Not a random split of one collection**, which measures only interpolation within that collection.
- **Per-subgroup results for every subgroup you can identify. Aggregate accuracy conceals exactly the failures that matter** — a model at 94% overall can be at 61% on a subgroup that is 5% of the data, and the aggregate will not move.
- **Performance under corruption:** blur, Gaussian and shot noise, JPEG compression, low light, simulated rain and fog, and the resolution changes of Module 08 §3. **These are cheap to generate synthetically and highly informative**, and standard corruption benchmarks exist.
- **A failure gallery of at least twenty cases, grouped by hypothesised cause** — which is the single most useful artefact for anyone deciding whether to rely on the system, and the one most often omitted.
- **Calibration** — a reliability diagram (CSCE 633 Module 12 §3), because §3 established that confidence is not distance from the boundary — **and latency measured on the hardware it will actually run on**, not on a development GPU.

THE HONEST CLAIM IS SCOPED

Not: 'our detector achieves 94% accuracy'. That sentence has no scope attached, so **it is not checkable and tells a reader nothing about whether to use the system.**

But: '94% on held-out data from the same three sites; 81% at a fourth site not used in training; 62% under heavy rain; worst identified subgroup 71%; systematically overconfident above 0.9; 28 ms per frame on the target device.'

That is longer, less impressive, and actually usable. It tells someone whether to deploy the system and under what conditions, which is the only purpose an evaluation has.

And it is the same discipline as every course in this program, in the form it takes when the input is the physical world: **state what you measured, state what you assumed, and never claim more than you established. In this course the assumption that needs stating is almost always about the capture conditions** — the camera, the lighting, the site — because that is what the model silently learned alongside the task.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Recht et al. — Do ImageNet Classifiers Generalize to ImageNet? (free)	The §1 replication. The measured gap, with the collection protocol described in enough detail to assess.
Hendrycks & Dietterich — Benchmarking Neural Network Robustness to Common Corruptions (free)	The corruption benchmark of §4, and the finding that clean accuracy does not predict corrupted accuracy.
Geirhos et al. — Shortcut Learning in Deep Neural Networks (free)	The §2 reference, with the documented cases and the detection strategies.
Madry et al. — Towards Deep Learning Models Resistant to Adversarial Attacks (free)	The §3 phenomenon treated rigorously, with the threat model stated explicitly — which is the part worth imitating.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Collect a second test set yourself,** differently, and report the gap against your original.
- 2 **Apply ten standard corruptions** and plot accuracy against severity.

- 3 **Rank three models on clean data and on corrupted data** and report whether the ranking changes.
- 4 **Evaluate per source** — per camera, site, or session — and look for the within/across gap.
- 5 **Occlude the object of interest** and report how much accuracy survives.
- 6 **Deliberately introduce a shortcut** into a training set and confirm the model takes it.
- 7 **Then detect it** with each of the methods in §2 and report which worked.
- 8 **Generate adversarial examples** for your model and measure the perturbation size needed.
- 9 **Plot a reliability diagram** and report where the model is overconfident.
- 10 **Write the scoped claim** for your own project in the form of §4's callout.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name five kinds of distribution shift and an example of each.
- 2 Which kind do vision systems meet constantly, and which causes harm?
- 3 What does the ImageNet replication establish?
- 4 Why can clean benchmark accuracy select the wrong model?
- 5 Give four documented shortcuts and say why each passed its test set.
- 6 Give five ways to detect a shortcut, and the cheapest reliable one.
- 7 What do adversarial examples establish, and what do they not?
- 8 Name five components of a real evaluation.
- 9 Write a scoped trust claim.